

Patricia Sorya

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Academic experience

2025 – 2027 **Postdoctoral researcher**, *Natural Sciences and Engineering Research Council of Canada (NSERC) fellowship* – University of Ottawa and Boston College, Ottawa and Boston.

Education

2021 – 2025 **Doctor of Philosophy**, *Pure mathematics*, Université du Québec à Montréal (UQAM), Montreal.

2019 – 2021 **Master of Science**, *Pure mathematics*, UQAM, Montreal.

2016 – 2019 **Bachelor of Science**, *Pure mathematics*, UQAM, Montreal.

2005 – 2010 **Doctor of Optometry**, Université de Montréal (UdeM), Montreal.

Research interests

Low dimensional topology

Knot theory

Topological data analysis

Publications

Sorya, Patricia. *Computing the Knot Floer complex of knots of thickness one*. Exp. Math. (2026) <https://doi.org/10.1080/10586458.2025.2603589>

Sorya, Patricia and Wakelin, Laura. *Effective bounds on characterising slopes for all knots*. Trans. Amer. Math. Soc. (to appear) [arXiv:2410.24209](https://arxiv.org/abs/2410.24209)

Sorya, Patricia. *Characterizing slopes for satellite knots*. Advances in Mathematics, vol. 450 (2024) <https://doi.org/10.1016/j.aim.2024.109746>

Chen C, Razavi R, Romano F, Niessen H, Sabokrohiyeh S, Konda A, Chevretils C, Arbour J, Rhéaume MA, Nissan R, Rojewski A, **Sorya P**, et al. *Hyperspectral Imaging for the Classification and Analysis of Drusen in Age-Related Macular Degeneration*. Invest. Ophthalmol. Vis. Sci. (2026) <https://doi.org/10.1167/iovs.67.3.10>

Nissan R, Chevretils C, **Sorya P**, et al. *Retinal phenotyping using spatial-spectral features derived from hyperspectral imaging*. Invest. Ophthalmol. Vis. Sci., vol. 65, 5953 (2024)

Sylvestre JP, Arbour JD, Rhéaume MA, Nissan R, Rojewski A, **Sorya P**, et al. *Evaluation of geographic atrophy, nascent geographic atrophy, and hyperreflective foci with hyperspectral retinal imaging*. Invest. Ophthalmol. Vis. Sci., vol. 65, 5953 (2024)

Nassar K, Niessen H, Arbour JD, Rhéaume MA, Nissan R, Rojewski A, **Sorya P**, et al. *Spatial-spectral characterization and mapping of labeled drusenoid deposits in non-neovascular age-related macular degeneration*. Invest. Ophthalmol. Vis. Sci., vol. 65, 5953 (2024)

Talks and conferences

- Invited speaker **On the number of shared Dehn surgeries between two knots.**
- American Mathematical Society 2026 Spring Eastern Sectional Meeting, Boston College, Boston, March 2026
 - Geometry and Topology seminar, University of Ottawa, Ottawa, November 2025
- Bounding non-integral non-characterizing Dehn surgeries.**
- Topology seminar, Stanford University, Stanford, April 2025
 - CIRGET Geometry and Topology seminar, UQAM, Montreal, January 2025
 - Topology seminar, Princeton University, Princeton, December 2024
 - Topology seminar, Georgia Institute of Technology, Atlanta, December 2024
 - Topology seminar, McMaster University, Hamilton, November 2024
 - Geometry and Topology seminar, University of Ottawa, Ottawa, November 2024
 - Geometry, Topology and Dynamics seminar, Boston College, September 2024
- Non-integral Dehn surgeries characterize composite knots.**
- Topology seminar, McMaster University, Hamilton, November 2024
 - Topology seminar, Dartmouth College, Hanover, September 2024
 - Topology seminar, Max Planck Institute for Mathematics, Bonn, July 2024
- A family of knots whose characterizing Dehn surgeries are the non-integral ones,** *Topology seminar*, University of Texas, Austin, October 2024.
- Characterizing slopes : Explicit bounds for satellite knots,** *Summer meeting of the Canadian Mathematical Society*, Ottawa, June 2023.
- Topological data analysis for data scientists: homology,** *Journal club of the Artificial Intelligence team*, Optina Diagnostics, Montreal, June 2023.
- Characterizing slopes for satellite knots,** *Winter meeting of the Canadian Mathematical Society*, Toronto, december 2022.
- Obstructions to the triangulation of manifolds,** *Geometric Topology Grad and Postdoc Seminar*, Standford University (online), February 2022.
- Speaker **Persistence modules and ribbon concordance,** *Geometry and Topology in Low Dimensions: Lasting Trends and Emerging Directions*, Budapest, June 2026 (upcoming).
- Persistence modules and ribbon concordance,** *Geometry & Topology seminar*, McMaster University, Hamilton, May 2026 (upcoming).
- Topological analysis of works for percussion ensembles,** *The Space Between V – Canadian Percussion Network*, McMaster University, Hamilton, May 2026 (upcoming).
- Knot Floer complex and characterizing Dehn surgeries of knots of thickness ≤ 2 ,** *New structures in low-dimensional topology*, Budapest, July 2024.
- Complexe de Floer de noeuds d'épaisseur 1,** *Colloque panquébécois de l'Institut des sciences mathématiques*, Montreal, June 2024.
- Classification of hyperspectral image data using persistence landscapes,** *Applications of Representation Theory in Topological Data Analysis & Geometric Invariant Theory*, Montreal, June 2024.
- Characterizing Dehn surgeries and composite knots,** *Colloque panquébécois de l'Institut des sciences mathématiques*, Sherbrooke, June 2023.
- Small hyperbolic links?,** *Groups Around 3-Manifolds*, Montreal, June 2023.

The Optina Diagnostics' Retinal Phenotyping Platform, 20^e Journée scientifique de l'École d'optométrie de l'Université de Montréal, Montreal, March 2023.

Characterizing slopes for satellite knots, *Winter school in singularities and low dimensional topology*, Budapest, January 2023.

Working groups **Speaker and attendee**, *Low dimensional topology working group of the CIRGET*, Montreal, 2019 – 2025.

Talks given:

- Computing the knot Floer complex, February-March 2025
- Explicit characterising slopes for all knots!, August 2024
- Intersection graphs and reducible fillings, February 2024
- Characterizing slopes for satellite knots, May 2023
- Signature of knots, April 2022
- A cross homomorphism for the Laudenbach exact sequence, February 2021
- Cyclic covers of knot complements, September 2020

Speaker, attendee, co-founder and organizer (rotating), *CIRGET Topology students reading group*, Montreal, 2021 – 2025.

Talks given:

- Introduction to the volume conjecture, February-March 2025
- Knot Floer homology, October 2023
- Dehn surgery, the fundamental group and $SU(2)$, after Kronheimer and Mrowka, June 2023
- A-polynomial: examples and properties, April 2023
- Character varieties: Culler-Shalen seminorms, February 2023
- Character varieties: ideal points and valuations, November 2022
- Character varieties: tree graphs and surfaces in 3-manifolds, October 2022
- Bordered Heegaard Floer homology (continued), October 2022
- Bordered Heegaard diagrams and their associated strands algebra, August 2022
- Twisted Alexander invariants, July 2022
- Heegaard Floer homology, Fall 2021

Speaker and attendee, *Topological data analysis working group*, UQAM and Université de Sherbrooke, online, 2021 – 2023.

Talks given:

- Persistence modules as sheaves, May 2022
- Topological data analysis to vectorize fMRI and hyperspectral scans, March 2022
- 2-parameter persistent homology, January 2022

Organizer **Member of the organizing committee**, *Colloque panquébécois de l'Institut des sciences mathématiques*, Montreal, May 2024.

Member of the organizing committee, *UQAM Graduate Students in Mathematics Seminar*, Montreal, 2019 – 2021.

Honors and awards

2025 – 2027 **Postdoctoral Fellowship**, *Natural Sciences and Engineering Research Council of Canada (NSERC)*, 140000\$.

2025 **Carl Herz Prize**, *Institut des sciences mathématiques (ISM)*, 5000\$.

2025 **Award for Excellence - Best student research**, *Faculty of sciences, UQAM*, 1000\$.

2025 **Scholarship for Outstanding PhD Candidates**, *ISM*, 7500\$.

2024 **ISM Graduate Scholarship**, *ISM*, 5000\$.

- 2021 – 2024 **Doctoral Research Scholarship**, *Fonds de recherche du Québec – Nature et technologies (FRQNT)*, 84000\$.
- 2020 – 2021 **Master's Research Scholarship**, *FRQNT*, 17500\$.
- 2019 – 2020 **Canada Graduate Scholarships – Master's Program**, *NSERC*, 17500\$.
- 2018 and 2019 **Undergraduate Student Research Award**, *NSERC*, 5625\$ and 6200\$.

Social engagement

- 2020 – 2025 **Student council member**, *Graduate Mathematics program committee*, UQAM, Montreal.
- Defining and aligning curriculum structure of graduate programs in mathematics and their objectives, in collaboration with council faculty members and the administration.
- April 2022 and 2023 **Invited speaker**, *MAT6221 History of mathematics*, UQAM, Montreal.
- Talk titled *Women mathematicians in geometry and topology*, highlighting the achievements of three female mathematicians of the 20th century.
 - Conference given as part of a course in the bachelor in mathematics education and the bachelor in mathematics programs.
- 2020 **Organizing committee and editing-revision**, *Project Femmes en maths*, online.
- Project to promote the contributions of women in mathematics
 - Maintenance of the website (<https://femmesenmaths.org/>), writing and editing-revision of publications, funding applications
- April 2019 **Invited panel member**, *Math Day for CEGEP Girls*, Montreal.

Teaching experience

- 2026 **Instructor**, *Department of Mathematics and Statistics – University of Ottawa*, Ottawa.
- MAT1362 Mathematical Reasoning and Proofs (Winter 2026)
- 2017 – 2023 **Lecturer and preceptor**, *School of Optometry – UdeM*, Montreal.
- OPM4801 Specialized clinical internship - Community clinic (2021 – 2023)
 - SCV2152 Vision sciences: Ocular dioptrics (Fall 2020)
 - OPM4701, OPM37011 Primary care clinical internship (2017 – 2021)
 - OPM6052 Advanced clinical optometry (Summer 2018)
- 2020 – 2021 **Teaching assistant**, *Département de mathématiques – UQAM*, Montreal.
- MAT2150 Analysis II (Fall 2021)
 - MAT2250 Group theory (Fall 2020 and 2021)
 - MAT2400 Geometry (Fall 2020)
 - MAT0339 General mathematics (Summer 2020)
 - MAT0344 Integral calculus (Winter 2020)

Professional experience

- 2021 – 2024 **Consultant, Data Analytics and Optometry**, *Optina Diagnostics*, Montreal.
- Analysis of clinical data acquired with Optina Diagnostics' ocular imaging technology, aimed at developing machine learning models to aid in the diagnosis of various systemic diseases
 - Designing research protocols and clinical documentation
 - Presenting results to collaborators and investors
 - Training staff at affiliated research sites

2017 – 2020 **Research Associate**, *Research Institute - McGill University Health Centre (RI-MUHC)*, Montreal.

- Oculovisual evaluation, diagnostic monitoring and therapy of patients enrolled in Phase II and III clinical studies
- Development of logistical processes for implementing research protocols, in collaboration with other professionals in the hospital setting

2010 – 2025 **Optometrist**, *member of the Ordre des optométristes du Québec*, no. 321032.

Languages

French

English